

Name _____ Teacher _____



GOSFORD HIGH SCHOOL

2014

HIGHER SCHOOL CERTIFICATE

ASSESSMENT TASK 1

MATHEMATICS – EXTENSION 1

Time Allowed - 60 minutes plus 5 minutes reading time

- Write using a black or blue pen. Black pen is preferred.
- Board approved calculators may be used.
- Answers to the multiple choice are to be done on the answer sheet provided.
- In questions 5-7, show relevant mathematical reasoning and/or calculations.

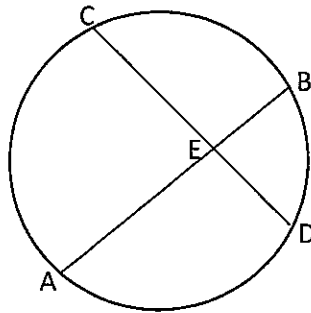
Section 1 Multiple choice	4 questions worth 1 mark each.	/4
Section 2 Question 5	Parametric treatment of the parabola	/12
Question 6	Polynomials	/12
Question 7	Mathematical Induction, Inequalities, Circle Geometry	/12
TOTAL		/40

SECTION 1: MULTIPLE CHOICE. Questions are worth 1 mark each. Answer on the multiple choice answer sheet provided.

- 1) Two lines have gradients of $m_1 = 2$ and $m_2 = \frac{1}{2}$ respectively. The angle between them to the nearest degree is:

A) 51° B) 90° C) 37° D) 72°

- 2) In the diagram below, $AB = \alpha$, $BE = \beta$, $CE = \gamma$ and $ED = \delta$



Which one of the following statements is true?

- A) $\gamma\delta = \alpha\beta$ B) $\frac{\gamma}{\delta} = \frac{\alpha-\beta}{\beta}$
 C) $\gamma(\gamma + \delta) = \beta(\alpha - \beta)$ D) $\gamma\delta = \beta(\alpha - \beta)$

- 3) A polynomial equation has roots α, β and γ where

$$\alpha + \beta + \delta = 3, \alpha\beta + \alpha\gamma + \beta\gamma = -2 \text{ and } \alpha\beta\gamma = 4$$

Which polynomial equation has the roots α, β and γ ?

- A) $x^3 + 3x^2 + 2x + 4$ B) $x^3 + 3x^2 + 2x - 4$
 C) $x^3 - 3x^2 - 2x - 4$ D) $x^3 - 3x^2 - 2x + 4$

- 4) The point C divides the interval from $A(-1,2)$ to $B(3,5)$ externally in the ratio 3:1. What is the x coordinate of C?

A) 5 B) -3 C) 2 D) 4

SECTION 2: Questions are worth 12 marks each. Answer on your own paper. Start each question on a new sheet of paper. All necessary working must be shown.

- 5) a) $P(2ap, ap^2)$ and $Q(2aq, aq^2)$ are 2 points on the parabola $x^2 = 4ay$.
Show that:
- i) the equation of the tangent at P on the parabola is $y = px - ap^2$. 2
- ii) the point of intersection, R, of the tangents at P and Q is $(a(p + q), apq)$ 2
- iii) the length of SP is $a(1 + p^2)$ where S is the focus of the parabola $x^2 = 4ay$. 2
- b) i) Find the equation of the chord of contact of the tangents to the parabola $x^2 = 8y$ from the point $P(4, -6)$. 2
- ii) Hence find the coordinates of the points of contact of these tangents with the given parabola. 2
- c) The variable point $(3t, 2t^2)$ lies on a parabola. Find the Cartesian equation for this parabola. 2

6) START A NEW PAGE

- a) $(x - 2)$ is a factor of the polynomial $P(x) = 2x^3 + x + a$. Find the value of a . **1**
- b) If α, β and γ are the roots of the equation $x^3 + 2x^2 - 11x - 12 = 0$, find
- i) $\alpha + \beta + \gamma$ **1**
- ii) $\alpha\beta + \alpha\gamma + \beta\gamma$ **1**
- iii) $\alpha\beta\gamma$ **1**
- iv) $(\alpha + 1)(\beta + 1)(\gamma + 1)$ **2**
- c) i) If 3 and -1 are 2 roots of $P(x) = x^3 - 8x^2 + 9x + 18$, express $P(x)$ in terms of three linear factors. **1**
- ii) Hence solve $P(x) < 0$ **2**
- c) Consider the equation $x^3 + 6x^2 - x - 30 = 0$. One of the roots of this equation is equal to the sum of the other two roots. Find the values of the three roots. **3**

7) START A NEW PAGE

- a)** Prove by mathematical induction that, for $n \geq 1$,

$$1 \times 5 + 2 \times 6 + 3 \times 7 + \dots + n(n+4) = \frac{1}{6}n(n+1)(2n+13) \quad \mathbf{3}$$

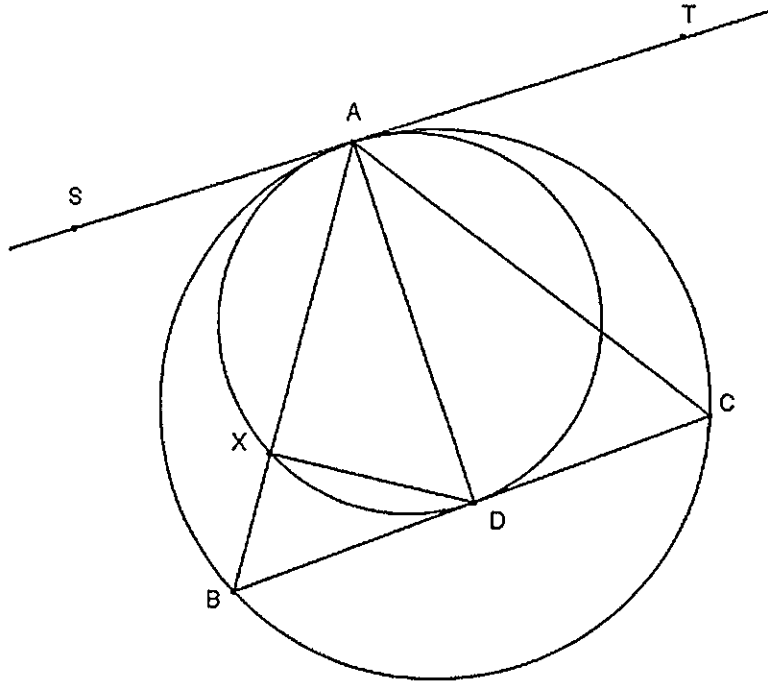
- b)** Prove by mathematical induction that

$$47^n + 53 \times 147^{n-1}$$

is divisible by 100 for all integers $n \geq 1$. 3

- c) Solve for x , $\frac{x^2-9}{x} > 0$

- d) In the diagram, ST is tangent to both the circles at A .
The points B and C are on the larger circle and the line BC is tangent to the smaller circle at D . The line AB intersects the smaller circle at X .



Copy the diagram onto your answer sheet.

- i) Explain why $\angle AXD = \angle ABD + \angle XDB$ **1**

- ii) Explain why $\angle AXD = \angle TAC + \angle CAD$ 1

- iii) Hence show that AD bisects $\angle BAC$ 2

SOLUTIONSSECTION 1

$$\tan \theta = \left| \frac{m_1 - m_2}{1 + m_1 m_2} \right|$$

$$= \left| \frac{2 - \frac{1}{2}}{1 + 1} \right|$$

$$= \frac{3/2}{2}$$

$$= \frac{3}{4}$$

$$\therefore \theta = 37^\circ \text{ --- (C)}$$

$$AE \cdot EB = CE \cdot ED$$

$$(\alpha - \beta)\beta = \gamma \cdot \delta \text{ --- (D)}$$

$$x^3 - 3x^2 - 2x - 4 = 0 \text{ --- (C)}$$

$$(-1, 2) \quad (3, 5)$$

$$3: -1$$

$$\frac{-1 \times -1 + 3 \times 3}{3 + -1} = \frac{1 + 9}{2}$$

$$= 5 \text{ --- (A)}$$

SECTION 2

$$1) a) P(2ap, ap^2) \quad Q(2aq, aq^2) \quad x^2 = 4ay$$

$$i) y = \frac{x^2}{4a}$$

$$\frac{dy}{dx} = \frac{2x}{4a}$$

$$= \frac{x}{2a}$$

$$\text{at } x = 2ap, \frac{dy}{dx} = \frac{2ap}{2a}$$

$$= p$$

$$\text{i.e. } m = p$$

$$\therefore y - y_1 = m(x - x_1)$$

$$y - ap^2 = p(x - 2ap)$$

$$y - ap^2 = px - 2ap^2$$

$$y = px - ap^2$$

$$ii) y = px - ap^2 \text{ --- (1)}$$

$$y = qx - aq^2 \text{ --- (2)}$$

$$px - ap^2 = qx - aq^2$$

$$px - qx = a(p^2 - q^2)$$

$$x(p - q) = a(p - q)(p + q)$$

$$x = a(p + q)$$

Sub into (1)

$$y = ap(p + q) - ap^2$$

$$= ap^2 + apq - ap^2$$

$$= apq$$

$\therefore$ point of intersection is $(a(p + q), apq)$

$$\begin{aligned}
 SP &= \sqrt{(2ap-0)^2 + (ap^2-a)^2} \\
 &= \sqrt{4a^2p^2 + a^2p^4 - 2a^2p^2 + a^2} \\
 &= \sqrt{a^2(p^4 + 2p^2 + 1)} \\
 &= \sqrt{a^2(p^2 + 1)^2} \\
 &= a(p^2 + 1)
 \end{aligned}$$

$$\begin{array}{lll}
 x^2 = 8y & 4a = 8 & x_0 = 4 \\
 & a = 2 & y_0 = -6
 \end{array}$$

$$i) \quad xx_0 = 2a(y+y_0)$$

$$4x = 4(y-6)$$

$$4x = 4y - 24$$

$$4x - 4y + 24 = 0$$

$$x - y + 6 = 0$$

$$ii) \quad x - y + 6 = 0 \quad \text{--- (1)}$$

$$x^2 = 8y \quad \text{--- (2)}$$

$$\text{from (2) } y = \frac{x^2}{8} \quad \text{--- (3)}$$

Sub (3) into (1)

$$x - \frac{x^2}{8} + 6 = 0$$

$$8x - x^2 + 48 = 0$$

$$x^2 - 8x - 48 = 0$$

$$(x-12)(x+4) = 0$$

$$x = 12, -4$$

$$\text{When } x=12, y=18$$

$$x = -4, y = 2$$

$\therefore$ points are (12, 18) and (-4, 2)

$$\therefore (3t, 2t^2)$$

$$x = 3t,$$

$$t = \frac{x}{3}, \quad y = 2t^2$$

$$y = 2 \cdot \frac{x^2}{9}$$

$$9y = 2x^2$$

$$x^2 = \frac{9}{2}y$$

$$\therefore a) \quad p(2) = 0$$

$$16 + 2 + a = 0$$

$$18 + a = 0$$

$$a = -18$$

$$b) \quad x^3 + 2x^2 - 11x - 12 = 0$$

$$i) \quad \alpha + \beta + \gamma = -\frac{b}{a} = -2$$

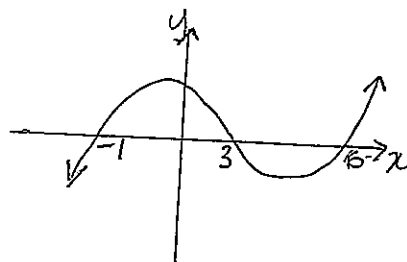
$$ii) \quad \alpha\beta + \alpha\gamma + \beta\gamma = \frac{c}{a} = -11$$

$$iii) \quad \alpha\beta\gamma = -\frac{d}{a} = 12$$

$$\begin{aligned}
 iv) \quad (\alpha+1)(\beta+1)(\gamma+1) &= \alpha\beta\gamma + \alpha\beta + \alpha\gamma + \beta\gamma + \alpha + \beta + \gamma + 1 \\
 &= 12 - 11 - 2 + 1 \\
 &= 0
 \end{aligned}$$

$$1) 1(x) = x^3 - 5x^2 + 7x + 10 \\ = (x-3)(x+1)(x-6)$$

ii)



$$P(x) < 0$$

when $x < -1$ or $3 < x < 6$

d) $x^3 + 6x^2 - x - 30 = 0$

let the roots be α , β and γ , where $\gamma = \alpha + \beta$

then

$$\alpha + \beta + (\alpha + \beta) = -6$$

$$2\alpha + 2\beta = -6$$

$$\alpha + \beta = -3 \quad \text{--- ①}$$

i.e. $\gamma = -3$

now $\alpha\beta\gamma = 30$

$$-3\alpha\beta = 30$$

$$\alpha\beta = -10 \quad \text{--- ②}$$

from ① $\beta = -3 - \alpha$

sub into ②

$$\alpha(-3 - \alpha) = -10$$

$$-3\alpha - \alpha^2 = -10$$

$$\alpha^2 + 3\alpha - 10 = 0$$

$$(\alpha + 5)(\alpha - 2) = 0$$

$$\alpha = -5, 2$$

$\therefore$ roots are $-5, -3, 2$

a) $1 \times 5 + 2 \times 6 + 3 \times 7 + \dots + n(n+4) = \frac{1}{6}n(n+1)(2n+15)$

Show true for $n=1$

$$\text{LHS} = 1(1+4) \\ = 5$$

$$\text{RHS} = \frac{1}{6} \cdot 1(1+1)(2+15) \\ = \frac{1}{6} \cdot 30 \\ = 5$$

$\therefore$ true for $n=1$

Assume true for $n=k$

i.e. $1 \times 5 + 2 \times 6 + 3 \times 7 + \dots + k(k+4) = \frac{1}{6}k(k+1)(2k+15)$

Prove true for $n=k+1$, if true for $n=k$

i.e. Prove $1 \times 5 + 2 \times 6 + \dots + k(k+4) + (k+1)(k+5) = \frac{1}{6}(k+1)(k+2)(2k+15)$

$$\text{LHS} = \frac{1}{6}k(k+1)(2k+15) + (k+1)(k+5)$$

$$= \frac{1}{6}(k+1)[k(2k+15) + 6(k+5)]$$

$$= \frac{1}{6}(k+1)[2k^2 + 15k + 6k + 30]$$

$$= \frac{1}{6}(k+1)(2k^2 + 21k + 30)$$

$$= \frac{1}{6}(k+1)(2k+15)(k+2)$$

$$= \text{RHS}$$

$\therefore$ true for $n=k+1$ if true for $n=k$

$\therefore$ true by mathematical induction for all $n \geq 1$

b) Prove $47^n + 53 \times 147^{n-1}$ divisible by 100 for $n \geq 1$

Show true for $n=1$

$$47 + 53 \times 147^0 = 100 \quad \therefore \text{true for } n=1$$

Assume true for $n=k$

i.e. $47^k + 53 \times 147^{k-1} = 100M$ where M is an integer

Prove true for $n=k+1$, if true for $n=k$

$$47^{k+1} + 53 \times 147^k = 47 \cdot 47^k + 53 \times 147 \cdot 147^{k-1}$$

$$\begin{aligned}
 &= 47 \cdot 47^n + 47(53 \times 147^{n-1}) + 100(53 \times 147^{n-1}) \\
 &= 47(47^k + 53 \times 147^{k-1}) + 100(53 \times 147^{k-1}) \\
 &= 47(100M) + 100(53 \times 147^{k-1}) \text{ from assumption} \\
 &= 100(47M + 53 \times 147^{k-1})
 \end{aligned}$$

$\therefore$ divisible by 100 when $n=k+1$, if divisible by 100 when $n=k$.

$\therefore$ by the principle of mathematical induction $47^n + 53 \times 147^{n-1}$ is divisible by 100 for all integers $n \geq 1$

i) $\frac{x^2-9}{x} > 0$

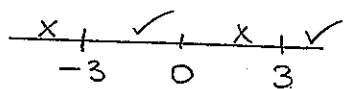
critical points

$x=0$

solve $\frac{x^2-9}{x} = 0$

$x^2-9=0$

$x = \pm 3$



test $x = -4$ $\frac{16-9}{-4} < 0$

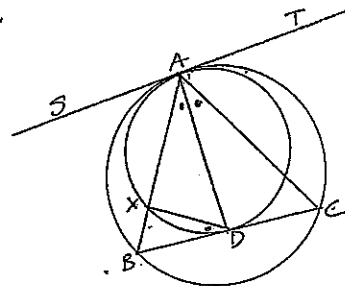
$x = -1$ $\frac{1-9}{-1} > 0$

$x = 1$ $\frac{1-9}{1} < 0$

$x = 4$ $\frac{16-9}{4} > 0$

$\therefore -3 < x < 0 \text{ or } x > 3$

d)



i) $\angle AXD$ is the exterior angle of $\triangle BD X$

$\therefore \angle AXD = \angle ABD + \angle XDB$ (exterior angle of $\triangle$ theorem)

ii) $\angle TAD = \angle TAC + \angle CAD$

$\angle AXD = \angle TAD$ (angle between tangent and chord equals the angle in the alternate segment)

$\therefore \angle AXD = \angle TAC + \angle CAD$

iii) $\angle CAD = \angle AXD - \angle TAC$ (from ii))

and $\angle XDB = \angle AXD - \angle ABD$ (from i))

but $\angle TAC = \angle ABC$ (angle between a chord and tangent equals the angle in the alternate segment)

$\therefore \angle CAD = \angle XDB$

but $\angle XDB = \angle XAD$ (angle between a chord and tangent equals the angle in the alternate segment)

$\therefore \angle BAD = \angle CAD$

$\therefore AD$ bisects $\angle BAC$